

22 A RMS voltage of supply = $12.0/\sqrt{2} = 8.49 \text{ V}$
p.d. across connecting wires = $8.49 - 7.00 = 1.49 \text{ V}$
Total resistance of 2 wires $R_T = 1.49/2.50 = 0.594 \Omega$,
Resistance of 1 wire = 0.297Ω
 $R = \rho L/A \rightarrow L = RA/\rho = 10.5 \text{ m}$

23 C In electric field